

ECO 303—INTERMEDIATE MICROECONOMIC THEORY

Fall 2016

Department of Economics
Stony Brook University

Instructor: Alejandro Melo Ponce

Lecture time: Tuesdays and Thursdays 4:00 PM–5:20 PM

Classroom: Melville Library E4330

Office hours: Thursdays 5:30PM–7:30 PM or by appointment, SBS N605

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Office hours: Monday 12:00 PM–1:00 PM (Xinwei), Friday 11:30 AM–12:30 PM (Chunwei); SBS Economics Lobby

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Course Description

The purpose of this course is to introduce the foundational tools of microeconomic analysis. As you know from your Principles of Economics class, Economics is the study of choice, i.e. how individuals or groups make decisions in a world where resources are limited. The overarching topics in this course are how consumers make choices both under certainty and uncertainty; how firms take decisions subject to technological constraints and in relation to the market structure they operate in and the organization of markets in the presence of complete and asymmetric information. To this end, the course aims to provide students with the economic and analytical tools to understand these topics.

The real world is very complex, so in order to gain some understanding, economists construct simple models to examine difficult economic questions. In this course I will present some basic models and show how they can be used to understand real world applications. Thus, this class will focus on developing techniques for analyzing economic questions. Whereas in your Principles of Economics class the focus was on learning some facts about economics, now we will focus in developing the tools of modern microeconomics. By the end of the course, you will become comfortable in applying those analytical tools in answering economic questions and also in applying them in more advanced economics classes. In order to develop these tools we will use mathematical concepts to do it. Modern Microeconomics uses math in order to communicate with precision and rigor its results, so this course will be an introduction to how modern economics is conducted. That being said, it is not my intention to make this course artificially hard or use more math than is required. This is a course in economics that relies on math, not a math course with some economic applications. Modern economics requires some math, so we are going to use it when necessary. However, to keep the focus on economics, I will try to give intuition and to stress the economic significance of the tools and results that we will cover.

Course Requirements

ECO 303 requires a previous course in Principles of Economics, which in Stony Brook is ECO 108. This course also requires a previous course at the level of calculus or higher. At Stony Brook the minimum requirement is either MAT 122, MAT 123 or AMS 151. Mathematics is a very useful tool for reasoning with precision, so as detailed in the introduction, we will use it regularly.

More precisely, the mathematics requirement for this course is *single variable calculus*. One of the first topics that we will cover in class is to learn some elementary techniques from *multivariate calculus*, but the techniques are just an extension of the single variable case, so this is the reason why there is no formal prerequisite for a multivariate calculus class. Everything that we need with respect to new mathematical tools will be developed in class.

Course Material

- *Microeconomic Theory: Basic Principles and Extensions* by Walter Nicholson and Christopher Snyder, 11th edition, Cengage Learning, 2011. This is the required textbook for the course. Each week you will have to do the required readings from this book. The 10th edition can also be used.
- *Intermediate Microeconomics with Calculus* by Hal R. Varian, 1st edition, W. W. Norton & Company, 2014. A more gentle exposition of the same topics. Recommended as supplementary reading.
- *Advanced Microeconomic Theory* by Geoffrey A. Jehle and Philip J. Reny, 3rd edition, Pearson, 2011. A more advanced treatment of Microeconomics for students that want to study the concepts in-depth.
- *Recommended. Essential Mathematics for Economic Analysis* by K. Sydsaeter and P. Hammond, Prentice-Hall 2008. Additional reading covering the mathematical concepts that we will use.
- I will assign some recent research articles and other readings to illustrate the models that we will cover in class and to provide real world applications. This material and my lecture slides will be posted on BlackBoard so it is your responsibility to check it regularly.

Recitations

Each week there will be a recitation section conducted by the Teaching Assistant. Each recitation will discuss past problems sets and elaborate and discuss the material from the lecture. Recitations sections will meet weekly beginning the second week of class. Students are strongly encouraged to attend their respective recitation sections because it provides an useful opportunity to work through problems, elaborate on concepts discussed in class and seek answers to questions.

Keeping up and course attendance

The course will move at a reasonable pace. Your priority every week should be to review the material of the week, including the lectures, the problem sets and the assigned reading from the textbook. Attending the lectures and the recitations is strongly advised so you can follow the material. By attending lectures you will be able to benefit from in-class presentations and discussions. You are also responsible for learning everything discussed in class whether these topics are in the textbook or not.

Rules and Regulations

During lectures, please keep your cell phones switched off and refrain from text messaging. Inattentive students whose actions are judged to be disruptive to class discussion will be asked to leave.

Grading Policy

Grading will be weighted as follows:

- Midterms (60%)
- Problem Sets (40%)

A letter grade based on the sum of the components above will be assigned as follows:

Letter Grade	Score Achieved
A	[93, 100]
A–	[90, 93)
B+	[87, 90)
B	[83, 87)
B–	[80, 83)
C+	[77, 80)
C	[73, 77)
C–	[70, 73)
D+	[67, 70)
D	[60, 67)
F	[0, 60)

Problem Sets

We will have 10–12 Problem Sets during the course. I will announce in class and in BlackBoard when a new problem set is posted. Problems sets are due *one week* after they are announced and you must deliver them at the end of the class. The graded problem sets will be returned in your recitation section, and any unclaimed problem sets will stay with the TA. The due date is strict. You are encouraged to discuss the problem sets in small groups of maximum five students. However, it is essential that you try to solve each question before you meet with your study group. It is indispensable that you attempt to solve the problems on your own if you wish to learn microeconomics. Each student will have to acknowledge the group members at the top of the solution to his/her problem set.

The problem sets can be downloaded from BlackBoard on the date they are announced. I will post answers on the day the problem set is due. Thus, late homework will not be accepted. I must receive your solutions at the end of the class on the day the problem set is due.

When computing your problem set average for grading purposes, I will drop the lowest score. Therefore, you can miss a problem set without repercussions.

Exams

You will have four exams, three midterms and one final exam. Each midterm is non-cumulative and covers the topics as outlined in table 1. We will hold the midterms in the usual lecture room. The dates for the midterms appear in table 1. If for any reason you anticipate that you won't be able to take the exam on that

date, please let me know as soon as possible. The final exam is cumulative and will cover all topics covered in the course. The final will be given during finals week, on Monday, December 19, from 2:15PM to 5:00PM, according to the Registrar's schedule.

I will drop the lowest scoring exam to compute your exam average. Therefore, if you are satisfied with your midterm grades, then you don't need to take the final. On the other hand if you missed one midterm or you just want to improve your exam average, you will have to take the final.

Tentative Course Outline

Table 1: Course Outline

Week	Class	Topic	Reading	Recitation
Week 1	8/30	Introduction, Preliminaries	NS 1,2,3	Rec 1
	9/1	Consumer Choice: Preferences and Utility functions		
Week 2	9/6	Labor day Weekend	NS 2,4	No Recitation
	9/8	Math Tools: Optimization and Constrained Optimization		
Week 3	9/13	Budget Sets, Demand and Choice	NS 5,6	Rec 2
	9/15	Budget Sets Demand and Choice		
Week 4	9/20	Demand functions and Elasticity		Rec 3
	9/22	Comparative Statics: Slutsky, Substitution and Income Effects		
Week 5	9/27	Labor Supply and Time		Rec 4
	9/29	Consumer Surplus		
Week 6	10/4	Midterm 1	NS 9,10,11	No Recitation
	10/6	Welfare, Taxes and Compensating Variation		
Week 7	10/11	Supply and Profit Maximization, Cost minimization	NS 12, 13	Rec 5
	10/13	Competitive Firms		
Week 8	10/18	Pareto Efficiency, Competitive Equilibrium and Edgeworth Box	NS 7	Rec 6
	10/20	Competitive Equilibrium: The First and Second Welfare Theorems		
Week 9	10/25	Choice under uncertainty	NS 7	Rec 7
	10/27	Choice under uncertainty: applications to financial markets		
Week 10	11/1	Midterm 2	NS 14	No Recitation
	11/3	Monopoly: Price Discrimination		
Week 11	11/8	Oligopoly	NS 15	Rec 8
	11/10	Strategic Behavior: Games		
Week 12	11/15	Dominated Strategies and Mixed Strategies	NS 8,18	Rec 9
	11/17	Asymmetric Information: Adverse Selection		
Week 13	11/22	Asymmetric Information: Moral Hazard		No Recitation
	11/24	Thanksgiving Weekend		
Week 14	11/29	Externalities and Public Goods	NS 8,18,19	Rec 10
	12/1	Second Degree Price Discrimination and Auctions		
Week 15	12/6	Auctions	NS 8,18	No Recitation
	12/8	Midterm 3		
	12/19	Final		

Special Needs

If you have a physical, psychological, medical or learning disability that may impact your course work, please contact Disability Support Services, 128 ECC Building (631) 632-6748. They will determine with you what accommodations are necessary and appropriate. All information and documentation is confidential.

Students who require assistance during emergency evacuation are encouraged to discuss their needs with their professors and Disability Support Services. For procedures and information go to the following web site: <http://www.ehs.sunysb.edu> and search Fire Safety and Evacuation and Disabilities.

Academic Integrity

Each student must pursue his or her academic goals honestly and be personally accountable for all submitted work. Representing another person's work as your own is always wrong. Faculty are required to report any suspected instances of academic dishonesty to the Academic Judiciary. For more comprehensive information on academic integrity, including categories of academic dishonesty, please refer to the academic judiciary website at http://www.stonybrook.edu/commcms/academic_integrity/index.html

Critical Incident Management

Stony Brook University expects students to respect the rights, privileges, and property of other people. Faculty are required to report to the Office of Judicial Affairs any disruptive behavior that interrupts their ability to teach, compromises the safety of the learning environment, or inhibits students' ability to learn. Faculty in the HSC Schools and the School of Medicine are required to follow their school-specific procedures.

Disclaimer

Subsequent changes can be made to any part of this syllabus when necessary. Any changes will be announced in class and on BlackBoard.